

Yoongon Kim

LinkedIn | GitHub

EDUCATION

Seoul National University

Bachelor of Science in Electrical and Computer Engineering | GPA 3.95/4.3

Seoul, South Korea

Mar. 2021 – Feb. 2027

University of California San Diego

Exchange Student

San Diego, CA

Sep. 2024 – Mar. 2025

EXPERIENCE

SNU VLSI Laboratory

Research Internship

Seoul, South Korea

Jun. 2024 – Present

- Developed the RelayGen framework to optimize inference efficiency, enabling a Small Language Model (SLM) to hand over complex reasoning to a Large Language Model (LLM) upon detecting a high-entropy token, validated across diverse benchmarks and architectures.
- Extended RelayGen to operate orthogonally with Speculative Decoding, designing a combined pipeline that achieved $2\times$ end-to-end inference speedup while maintaining model output quality.

UCSD Existential Robotics Laboratory

Research Internship

San Diego, CA

Sep. 2024 – Mar. 2025

- Created tutorials for robotics beginners, covering essential mobile robotics algorithms—including localization, mapping, motion planning, and control—to support structured and self-paced learning.
- Built a PyQt-based GUI for PyBullet simulations, enabling intuitive visualization of algorithm behaviors and real-time experimentation.

PROJECTS

LowerBound - LLM-Powered Crypto Risk Diversifier | Python, TypeScript, Docker, REST API, Git Oct 2025

- Built an LLM-driven chat interface and risk-engine pipeline that generates user-tailored crypto portfolios by interpreting investment goals and risk tolerance.
- Implemented a risk engine that runs crash-scenario stress tests on historical data to validate the robustness of generated portfolios.
- Generated intuitive visual explanations by converting complex risk metrics and allocation rationales into diagrams and graphs.

SNU FastMRI Challenge | PyTorch, OpenCV, Wandb, Git

Jul. – Aug. 2024

- Built a deep-learning model that reconstructs fully-sampled k-space MRI data from under-sampled measurements in the FastMRI dataset.
- Designed a Mixture-of-Experts framework using four FI-VarNet sub-models, improving SSIM by 21% over the baseline.
- Performed ablation to show block-wise attention had minimal effect, then removed it to fit the 8GB GPU limit, freeing 2GB VRAM for deeper U-Nets.
- Ranked 4th out of 127 teams on the public leaderboard.

Donggle - Pet Robot for Elderly | MediaPipe, OpenCV, Raspberry Pi

Jan. 2022

- Developed a Raspberry Pi-based pet robot that detects emergency events (falls, abnormal movements) in real time using MediaPipe and OpenCV.
- Achieved 5th place out of 20 teams at the Seoul Hardware Hackathon.

HONORS & AWARDS

Semiconductor Specialized Program Scholarship

Awarded to high-achieving ECE students based on academic excellence and research potential

Seoul National University

Aug. 2025

Korea–U.S. Advanced Tech Exchange Scholarship

\$9,000 government-funded award for STEM students in U.S. academic exchange programs

Ministry of Trade, Industry and Energy

Sep. 2024

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, HTML/CSS

Frameworks / Libraries: React Native, React, PyQt, PyBullet, MediaPipe

Developer Tools / Platforms: Git, Google Cloud Platform, VS Code, Visual Studio, PyCharm, Wandb, Firebase